

Recognition does not bind

Evidence-binding failures in language-model agents: findings from a forensic transcript corpus, and a proposed Recall–Verify–Bind architecture

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ABSTRACT

Failures of large language model agents are commonly attributed to missing knowledge. Analysis of a corpus of investigation reports, session transcripts, and reconstructed case studies drawn from real agent interactions suggests a distinct and understudied failure mode: the correct information was available — frequently already retrieved, acknowledged, or correctly explained by the agent itself — yet failed to constrain the agent's subsequent behavior. This phenomenon is termed **recognition without binding**. Reliable action is decomposed into three stages: information must be *available*, it must be *inspected*, and the inspected evidence must become *binding* on future decisions. Current research concentrates on the first stage and, to a lesser degree, the second; the third is largely unexamined. In one manually audited sample, roughly two-thirds of externally verifiable claims followed visible inspection; a meaningful minority did not, including completion claims made in the absence of any deliverable. A second sample on different material produced comparable inspection rates. An earlier estimate of a seventy-five percent overall failure rate was rejected as unsupported by the available evidence. On these findings, a **Recall–Verify–Bind** architecture is proposed — carrying unresolved obligations across sessions, requiring inspection before externally verifiable claims, and withholding declarations of success until objective evidence of completion exists — together with a four-condition evaluation designed to test whether deterministic binding reduces these failures without imposing prohibitive friction.

1. Background: what happened

The investigation began with a conventional assumption: that agent errors reflect gaps in knowledge, and that the remedy is larger models, better retrieval, or more memory. The accumulated evidence pointed elsewhere. Across dozens of documented sessions, agents repeatedly held — and in many cases had already located, quoted, or correctly summarized — the exact information needed for the decision at hand, and then acted as though that information did not exist.

The pattern recurred in recognizable forms. An agent claimed that credentials or integrations were missing when the necessary configuration was already present in its environment, defaulting to requests for additional setup rather than inspecting what it had. An agent declared work complete when no deliverable had been produced. An agent quoted a rule it had just been taught and violated that rule within the same conversation. In the most striking cases, governance mechanisms constructed specifically to prevent such failures were themselves subsequently ignored. The common denominator was not ignorance but a **failure to verify before claiming**, and a failure of recognized evidence to govern the next action.

2. Why: a three-stage account of reliable action

The central thesis is that recognition and governance are separable. An agent can acknowledge a correction, retrieve a document, or summarize the decisive evidence, and still proceed unconstrained by it. Reliable behavior requires three stages, each of which can fail independently:

1. **Availability.** The information exists somewhere reachable — in context, in memory, on disk, in an integration.
2. **Inspection.** The agent actually examines that information before acting on assumptions about it.
3. **Binding.** The inspected evidence becomes constraining: it forecloses actions inconsistent with it.

Most current reliability work targets the first stage (scale, memory, context length) and some the second (retrieval, search prompting). The third stage — whether what an agent knows can constrain what it does next — is where a substantial share of observed real-

world failures appear to occur, and where almost no systematic study exists. This reframes a portion of what is loosely called *hallucination* as a governance problem rather than a knowledge problem.



Central thesis

Recognition does not bind. Recognizing the truth is not the same as being governed by it: an agent can acknowledge a correction, retrieve a document, or summarize the decisive evidence, and still proceed to ignore it in its next decision.

3. Method

The corpus comprises dozens of investigation reports, raw transcripts, and reconstructed case studies from real agent interactions. Rather than coding every error as hallucination, each incident was audited against a three-way classification: was the relevant information *unavailable*, *overlooked* (available but never inspected), or *ignored* (inspected, even articulated, and then not acted upon)? Claims made by agents were checked against externally verifiable ground truth where the transcript permitted — environment state, produced artifacts, prior turns in the same session.

The audit was deliberately adversarial toward its own hypothesis. Counterexamples were documented rather than discarded: some failures genuinely resulted from unavailable information, and others were legitimate capability limitations. Quantitative claims that could not be reconstructed from the underlying data were withdrawn — most notably an early claim of a seventy-five percent overall failure rate, which a subsequent audit concluded could not be justified and which is not asserted here.

4. Findings

The defensible findings are narrow and sample-specific. In one manually reviewed dataset, roughly **two-thirds of externally verifiable claims** showed visible inspection before they were made; the remaining minority did not. A subset of completion claims lacked any evidence that the promised work had been performed. A second sample, drawn from

different material, produced similar inspection rates. The significance is not the precise percentages but their stability: the failures are systematic enough to measure, not isolated anecdotes.

Qualitatively, the ignored-evidence cases share a signature. Effort migrates from inspection to explanation: the agent constructs increasingly elaborate accounts of why it cannot proceed while never performing the single cheap check that would resolve the uncertainty. The cost asymmetry is stark — in the recurring integration example, verification was one command away — which suggests the failure is not a resource limitation but an ungoverned default toward assertion over inspection.

5. What next: Recall, Verify, and Bind

The proposed research program shifts the question from *does the model know more?* to *can inspected information be made binding?* The candidate mechanism is a deterministic evidence-binding layer, distinct from memory or reminders, with three obligations:

Recall carries unresolved obligations across sessions, so that corrections and commitments survive context boundaries rather than expiring with the conversation. **Verify** requires inspection before any externally verifiable factual claim — an agent may not assert the state of an environment it has not examined. **Bind** prevents a declaration of success until objective evidence exists that the requested work was actually completed.

6. Proposed evaluation

The architecture is treated as an experimental hypothesis, not a conclusion. Four conditions isolate whether binding — as opposed to instruction or memory — reduces the measured failures, and at what cost in friction and refused-but-legitimate work:

Condition	Mechanism	Question it isolates
C1 · Baseline	Unmodified agent	Prevalence of binding failures absent intervention
C2 · Prompted	Verification instructions only	Whether telling the agent to verify suffices
C3 · Memory	Persistent memory augmentation	Whether remembering is the missing capacity
C4 · Binding	Deterministic evidence binding (RVB)	Whether enforcement reduces failures without prohibitive friction

Primary outcome measures follow the audit method of §3: inspection rate before externally verifiable claims, rate of unsupported completion claims, and rate of rule violations following acknowledged corrections — alongside secondary measures of task throughput and false-blocking.

7. Significance

The contribution is a reframing of the reliability problem. Memory, retrieval, reasoning, and planning are treated in current research as the central obstacles to trustworthy agents; this work identifies an orthogonal dimension — **governance** — and supplies evidence that failures on that dimension are recurrent, measurable, and unaddressed by making models more knowledgeable. If the hypothesis survives the proposed evaluation, a principal obstacle to trustworthy AI is not the acquisition of knowledge but the enforcement of it: ensuring that once evidence is recognized, the system can no longer act as though it never existed.